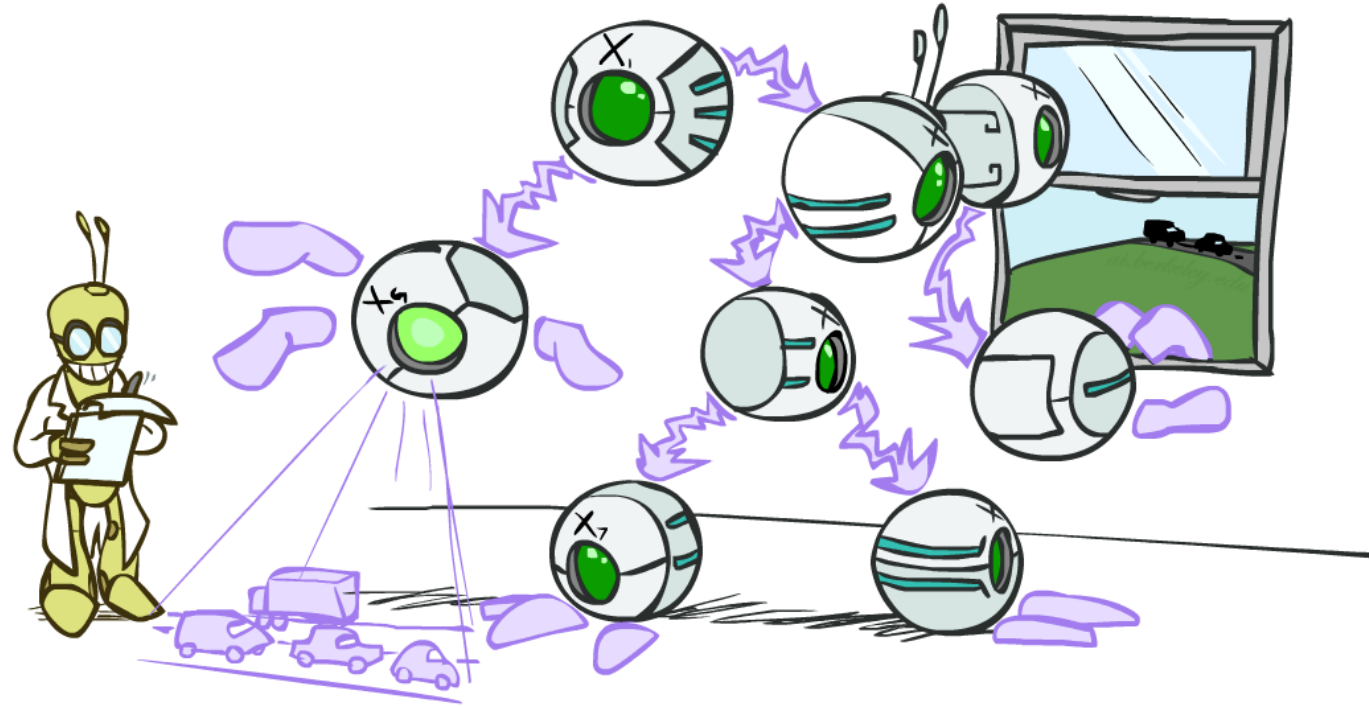


Bayes Nets: Exact Inference



AIMA Chapter 14.4, PRML Chapter 8.4

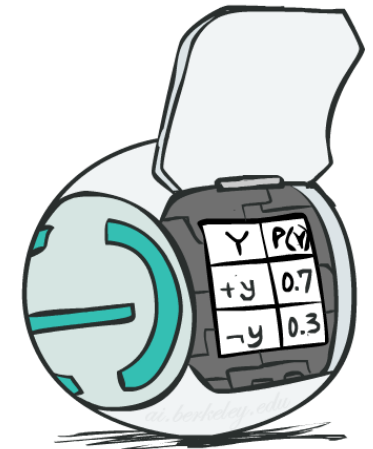
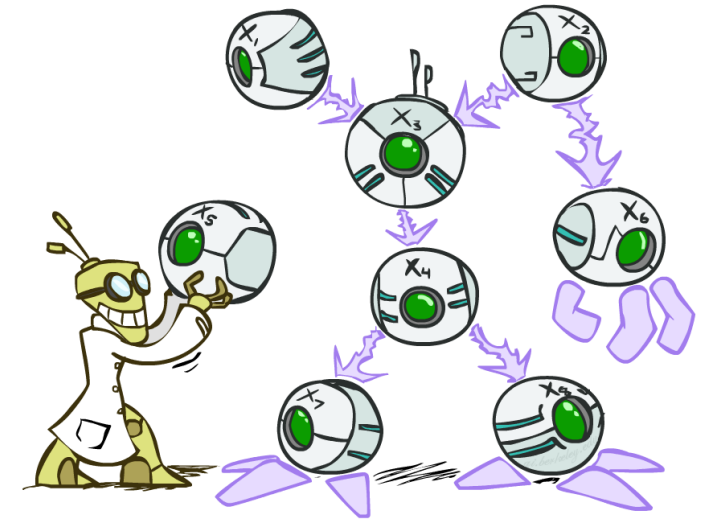
Bayes' Net Representation

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

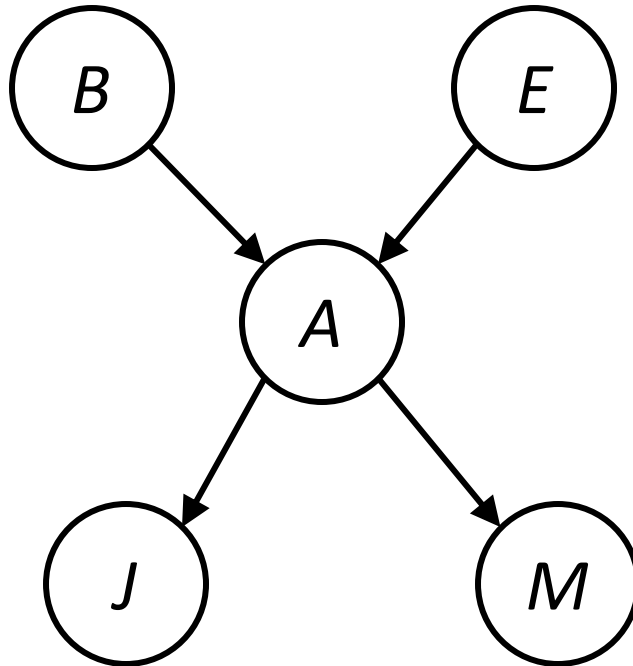
- Bayes' nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



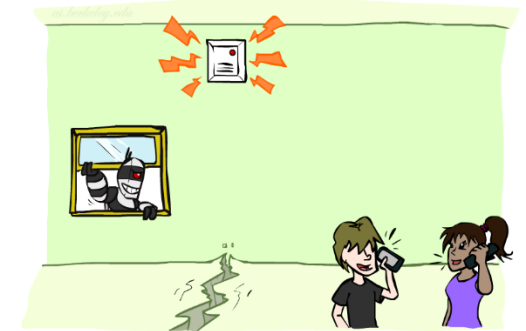
Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

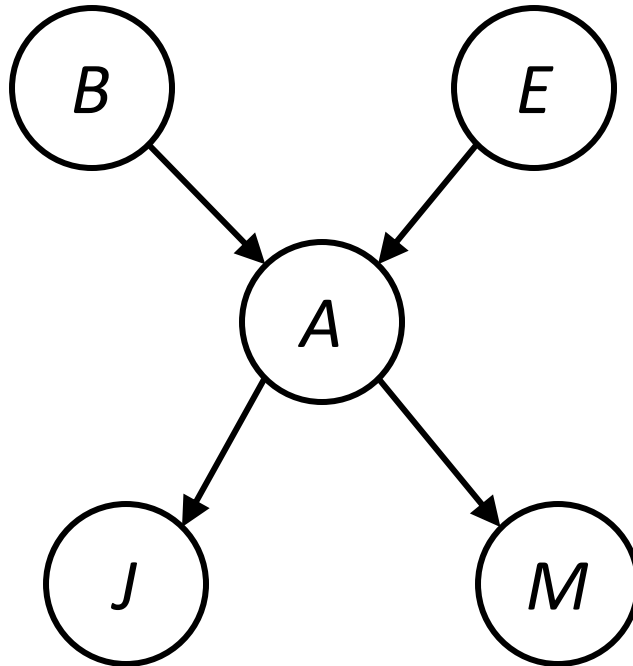
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m) =$$

$$P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) =$$

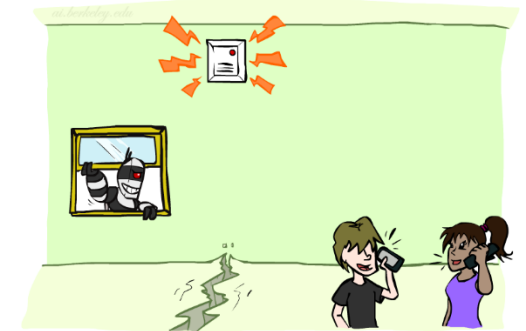
Example: Alarm Network

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+b	0.001
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-a	-m	0.99



A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

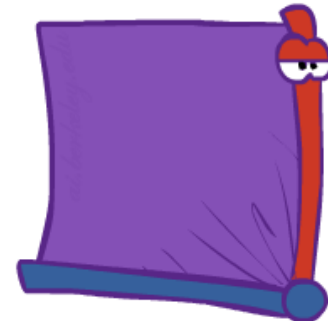
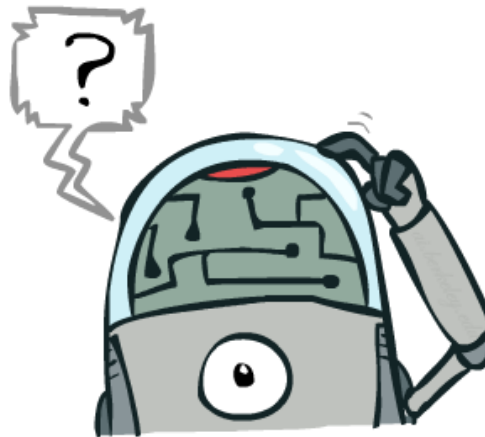
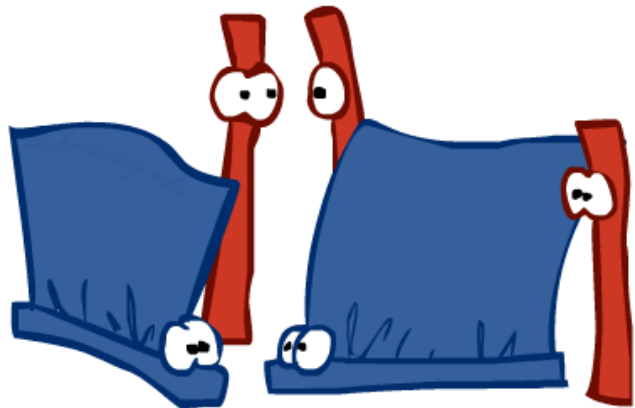
$$\begin{aligned}
 &P(+b, -e, +a, -j, +m) = \\
 &P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) = \\
 &0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7
 \end{aligned}$$

Probabilistic Inference

- Enumeration (exact, exponential complexity)
- Variable elimination (exact, worst-case exponential complexity, often better)
- Inference is NP-complete
- Sampling (approximate)

Inference

- Inference: calculating some useful quantity from a probability model (joint probability distribution)
- Examples:
 - Posterior marginal probability
 - $P(Q|e_1, \dots, e_k)$
 - E.g., what disease might I have?
 - Most likely explanation:
 - $\operatorname{argmax}_q P(Q=q|e_1, \dots, e_k)$
 - E.g., what did he say?



Inference by Enumeration

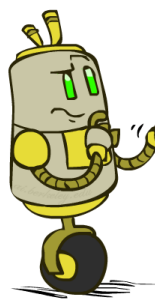
General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $\left. \begin{array}{l} X_1, X_2, \dots, X_n \\ \text{All variables} \end{array} \right\}$

We want:

$$P(Q|e_1 \dots e_k)$$

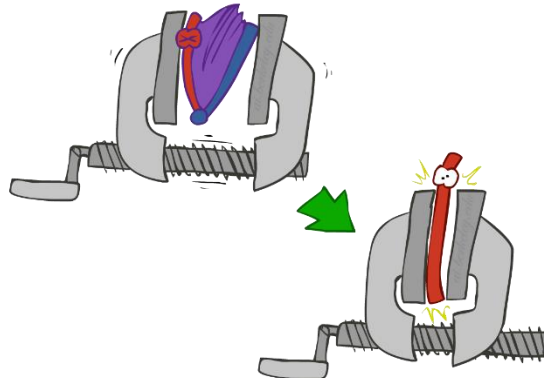
- Step 1: Select the entries consistent with the evidence



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2 0.15

- Step 2: Sum out H to get joint of Query and evidence



- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

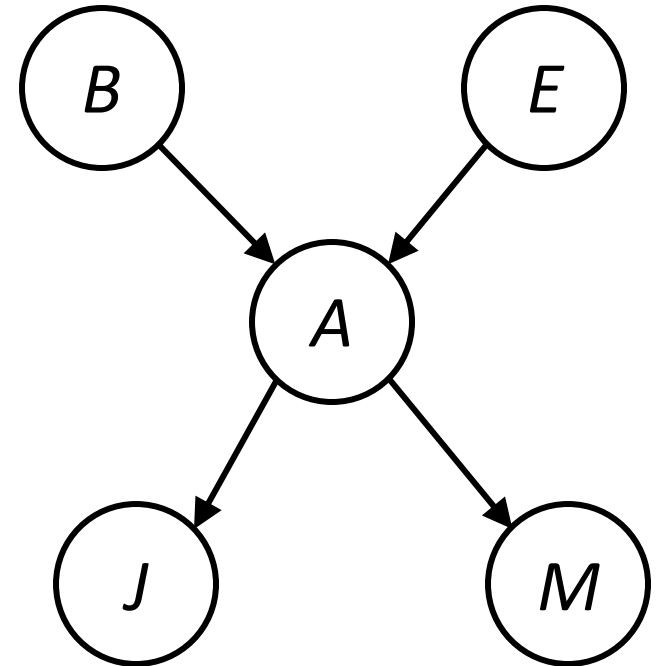
$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots, X_n}$$

Inference by Enumeration in Bayes Net

- The joint distribution can be computed from a BN by multiplying the conditional distributions
- Then we can do inference by enumeration

$$\begin{aligned} P(B \mid +j, +m) &\propto_B P(B, +j, +m) \\ &= \sum_{e,a} P(B, e, a, +j, +m) \\ &= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a) \end{aligned}$$

- Problem: sums of **exponentially many** products!

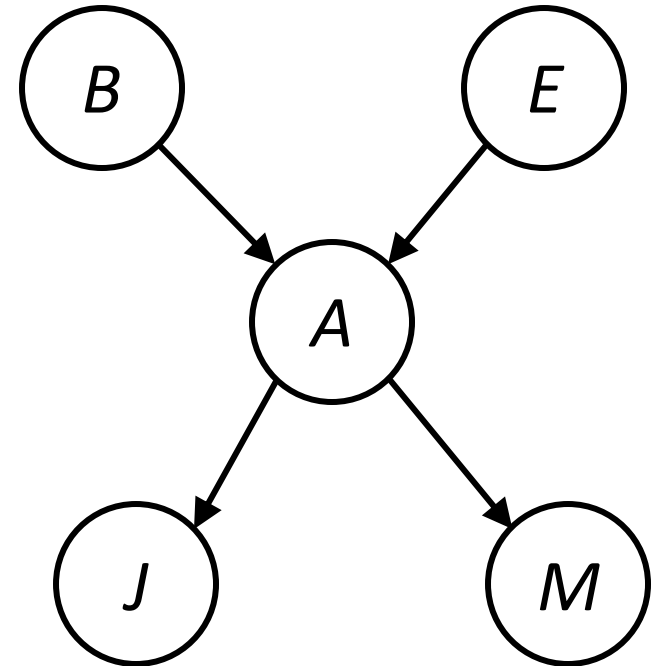


Inference by Enumeration in Bayes Net

$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a)$$



$$\begin{aligned} &= P(B)P(+e)P(+a|B, +e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B, +e)P(+j|-a)P(+m|-a) \\ &\quad P(B)P(-e)P(+a|B, -e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B, -e)P(+j|-a)P(+m|-a) \end{aligned}$$

Lots of repeated subexpressions!

Can we do better?

- Consider $uwy + uwz + uxy + uxz + vwy + vwz + vxy + vxz$
 - 16 multiplies, 7 adds
 - Lots of repeated subexpressions!
- Rewrite as $(u+v)(w+x)(y+z)$
 - 2 multiplies, 3 adds

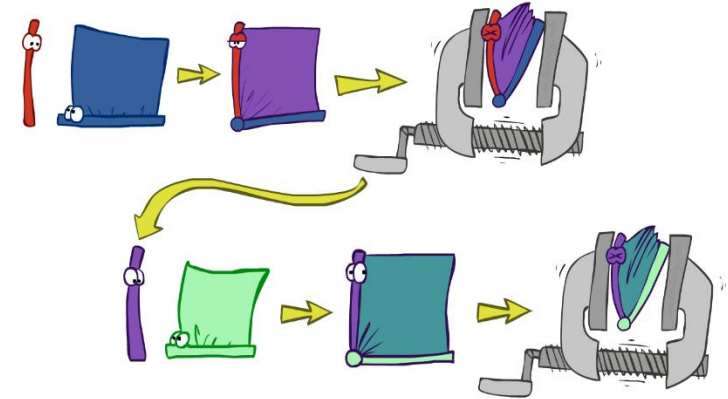
Can we do better?

- $\sum_{e,a} P(B) P(e) P(a | B, e) P(j | a) P(m | a)$
- $= P(B)P(e)P(a | B, e)P(j | a)P(m | a) + P(B)P(\neg e)P(a | B, \neg e)P(j | a)P(m | a)$
 $+ P(B)P(e)P(\neg a | B, e)P(j | \neg a)P(m | \neg a) + P(B)P(\neg e)P(\neg a | B, \neg e)P(j | \neg a)P(m | \neg a)$

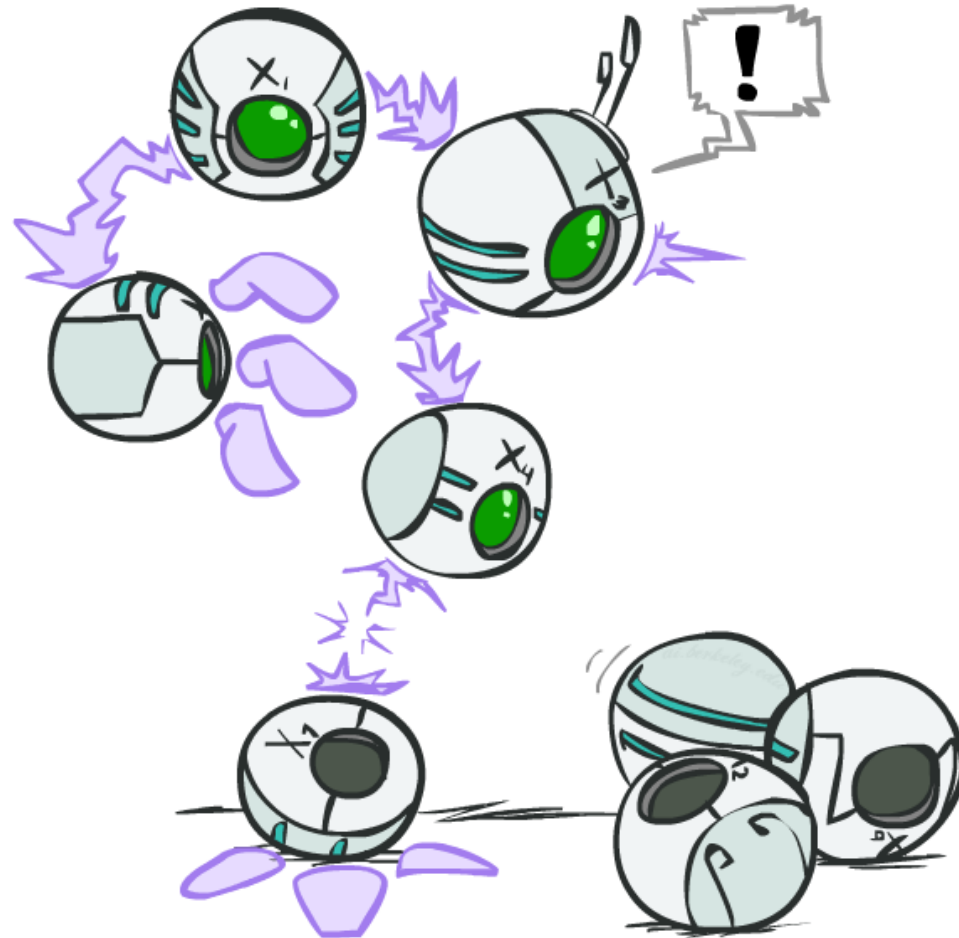
Lots of repeated subexpressions!

Variable elimination: The basic ideas

- Move summations inwards as far as possible
 - $P(B \mid j, m) = \alpha \sum_{e,a} P(B) P(e) P(a \mid B, e) P(j \mid a) P(m \mid a)$
 - $= \alpha P(B) \sum_e P(e) \sum_a P(a \mid B, e) P(j \mid a) P(m \mid a)$
- Do the calculation from the inside out
 - I.e., sum over a first, the sum over e
 - Problem: $P(a \mid B, e)$ isn't a single number, it's a bunch of different numbers depending on the values of B and e
 - Solution: use arrays of numbers (of various dimensions) with appropriate operations on them; these are called **factors**



Operations on Factors



Factors

- A **factor** is a multi-dimensional array to represent $P(Y_1 \dots Y_N \mid X_1 \dots X_M)$
 - If a variable is assigned (represented with lower-case), its dimension is missing (selected) from the array

- Joint distribution: $P(X,Y)$

- Entries $P(x,y)$ for all x, y
- Sums to 1

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Selected joint: $P(x,Y)$

- A slice of the joint distribution
- Entries $P(x,y)$ for fixed x , all y
- Sums to $P(x)$

$$P(\text{cold}, W)$$

T	W	P
cold	sun	0.2
cold	rain	0.3

Factors

- A **factor** is a multi-dimensional array to represent $P(Y_1 \dots Y_N \mid X_1 \dots X_M)$
 - If a variable is assigned (represented with lower-case), its dimension is missing (selected) from the array

- Single conditional: $P(Y \mid x)$

- Entries $P(y \mid x)$ for fixed x , all y
- Sums to 1

$$P(W \mid cold)$$

T	W	P
cold	sun	0.4
cold	rain	0.6

- Family of conditionals:

$$P(X \mid Y)$$

- Multiple conditionals
- Entries $P(x \mid y)$ for all x, y
- Sums to $|Y|$

$$P(W \mid T)$$

T	W	P
hot	sun	0.8
hot	rain	0.2
cold	sun	0.4
cold	rain	0.6

$$P(W \mid hot)$$

$$P(W \mid cold)$$

Factors

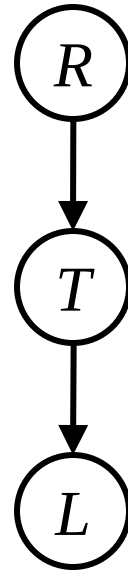
- A **factor** is a multi-dimensional array to represent $P(Y_1 \dots Y_N \mid X_1 \dots X_M)$
 - If a variable is assigned (represented with lower-case), its dimension is missing (selected) from the array
- Specified family: $P(y \mid X)$
 - Entries $P(y \mid x)$ for fixed y , but for all x
 - Sums to ... who knows!

$P(rain T)$			
T	W	P	
hot	rain	0.2	} $P(rain hot)$ $P(rain cold)$
cold	rain	0.6	

Example: Traffic Domain

- Random Variables

- R: Raining
- T: Traffic
- L: Late for class!



$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

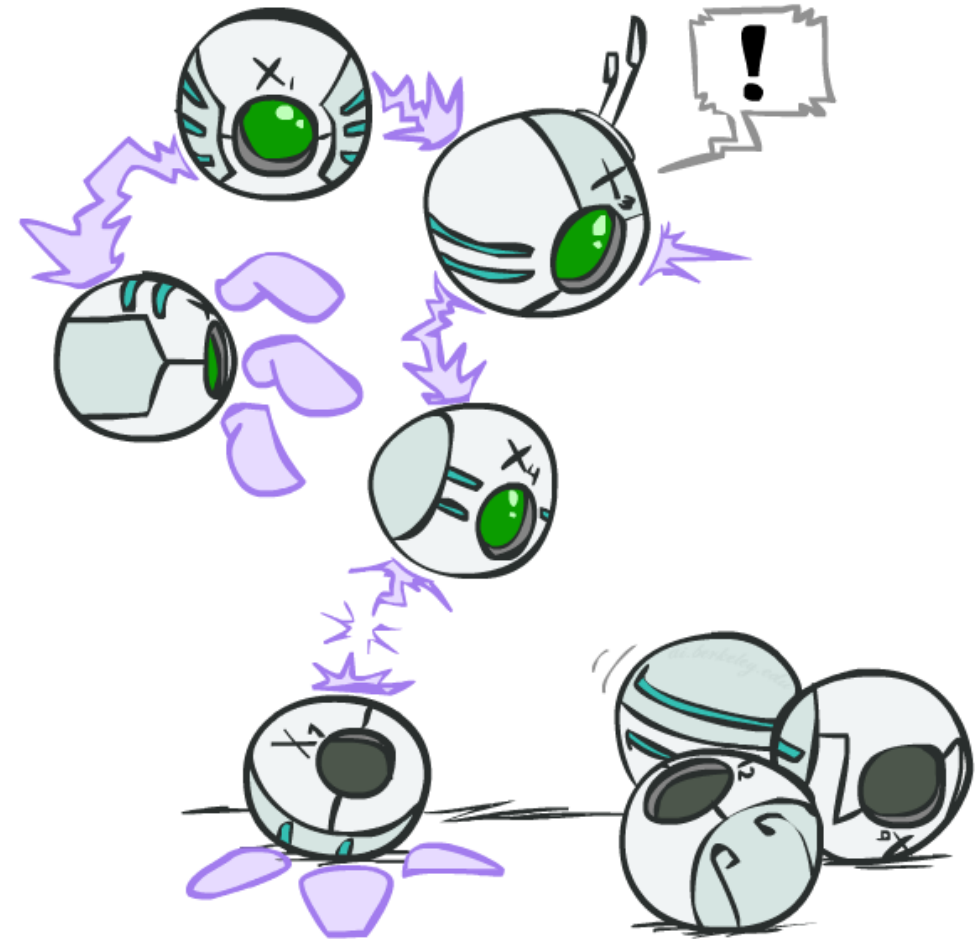
Running Example: Traffic Domain

- Initial factors are local CPTs (one per node)

$P(R)$		$P(T R)$			$P(L T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
-r	0.9	+r	-t	0.2	+t	-l	0.7
		-r	+t	0.1	-t	+l	0.1
		-r	-t	0.9	-t	-l	0.9

- Any known values are selected
 - E.g. if we know $L = +\ell$, the initial factors are

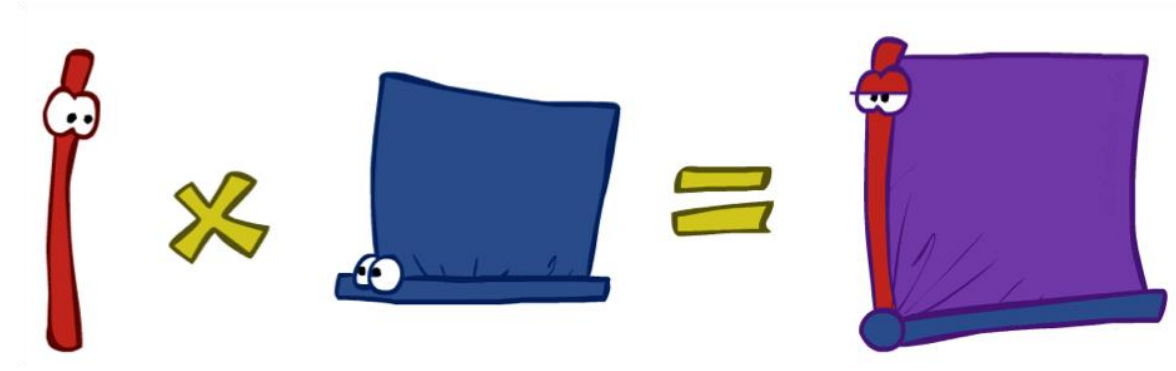
$P(R)$		$P(T R)$			$P(+\ell T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
-r	0.9	+r	-t	0.2	-t	+l	0.1
		-r	+t	0.1			
		-r	-t	0.9			



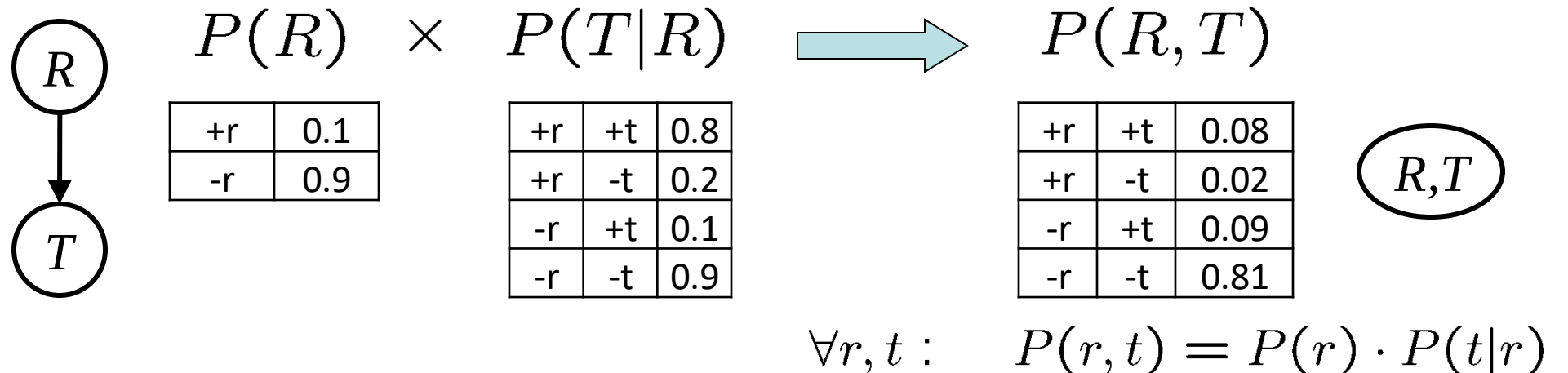
Operation 1: Join Factors

- First basic operation: **joining factors**

- Just like a database join
- Given multiple factors, build a new factor over the union of the variables involved
- Each entry is computed by pointwise products



- Example:



Operation 2: Eliminate

- Second basic operation: **eliminating a variable**
 - Take a factor and sum out (marginalize) a variable
- Example:

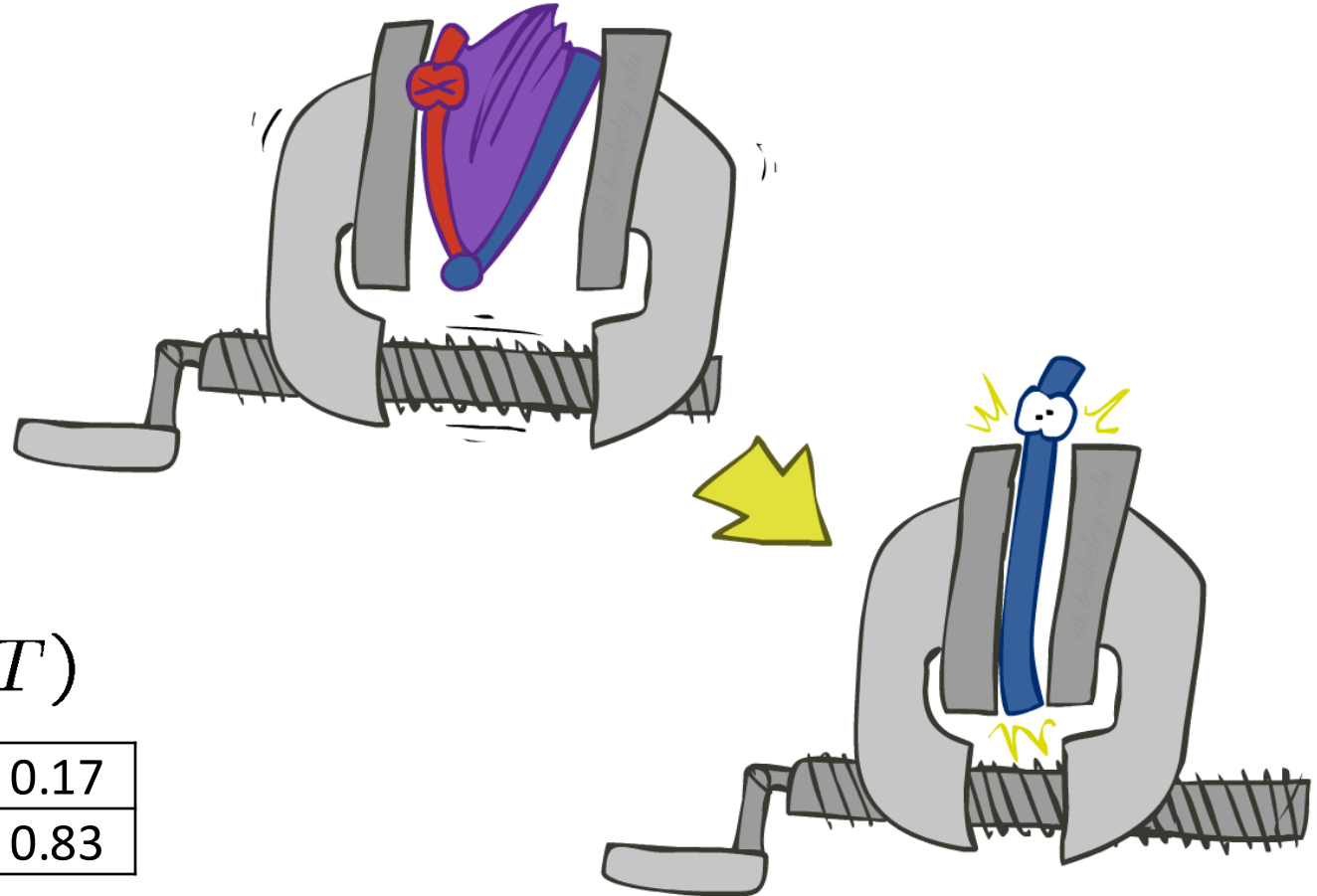
$$P(R, T)$$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

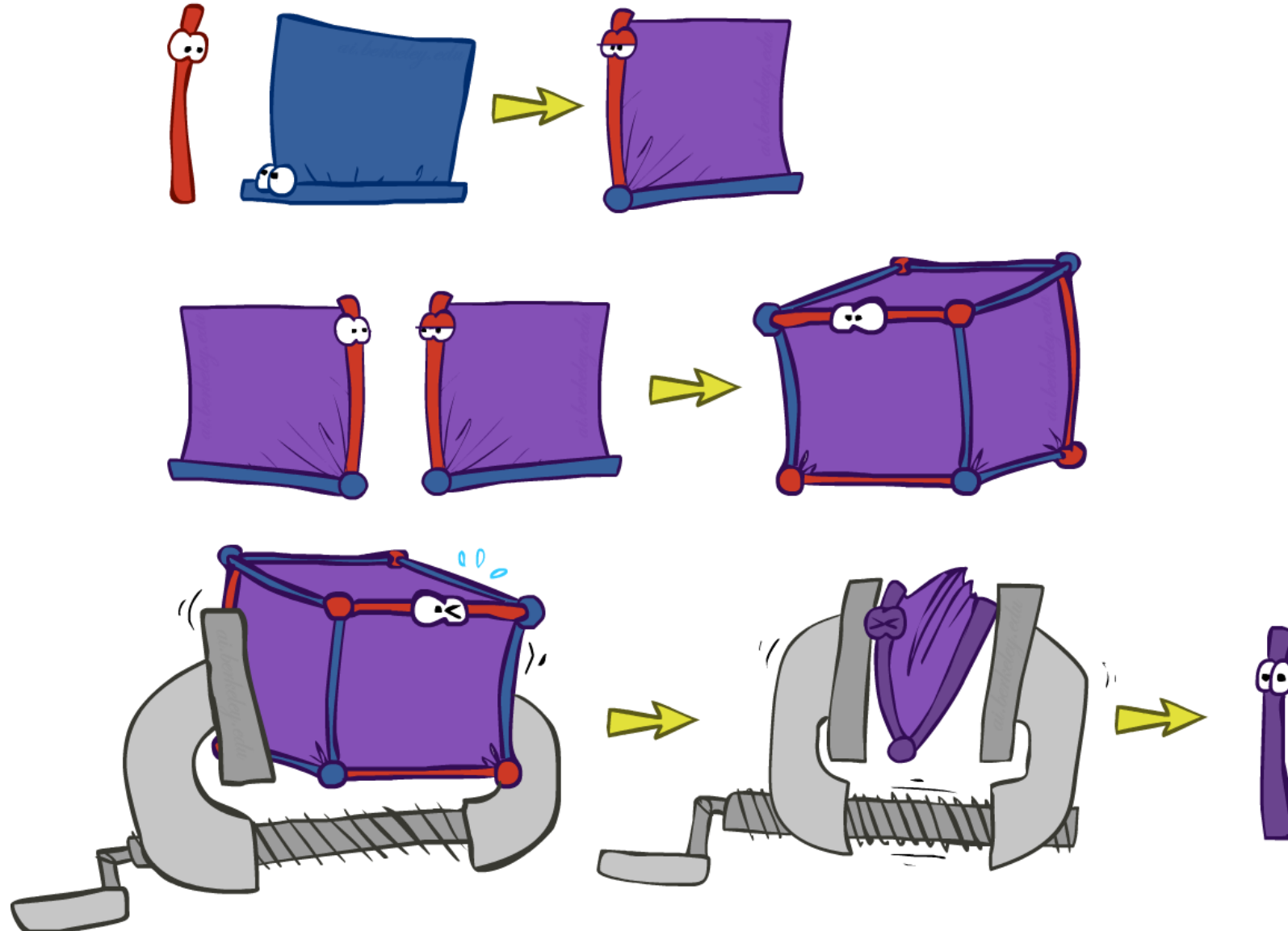
sum R


$$P(T)$$

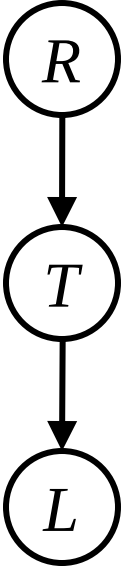
+t	0.17
-t	0.83



Inference by Enumeration in BN = Multiple Join + Multiple Eliminate



Computing $P(L)$: Multiple Joins



$P(R)$

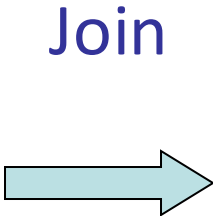
+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

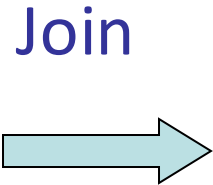
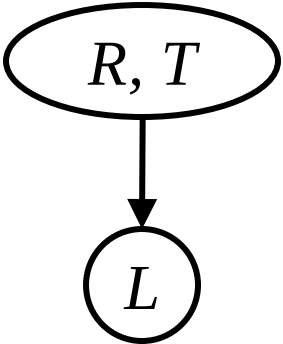


$P(R, T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

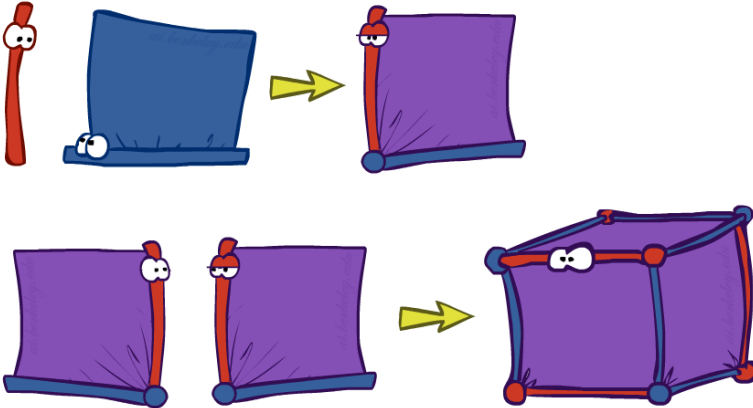
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9



$P(R, T, L)$

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729



Computing $P(L)$: Multiple Elimination

$P(R, T, L)$

R, T, L			
+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Sum out R

$P(T, L)$

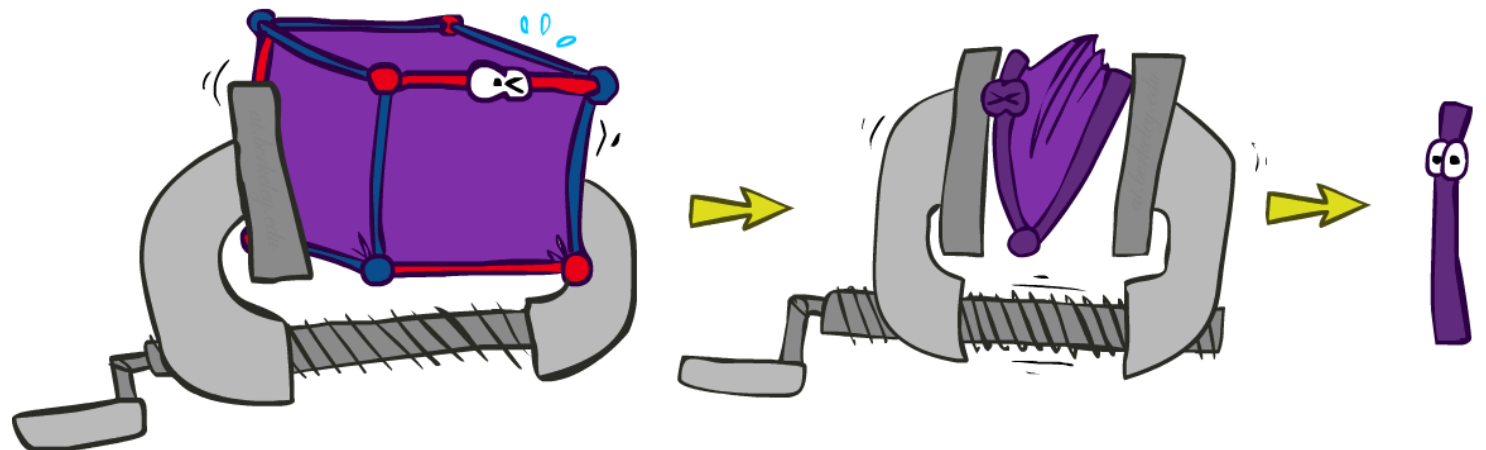
T, L		
+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T

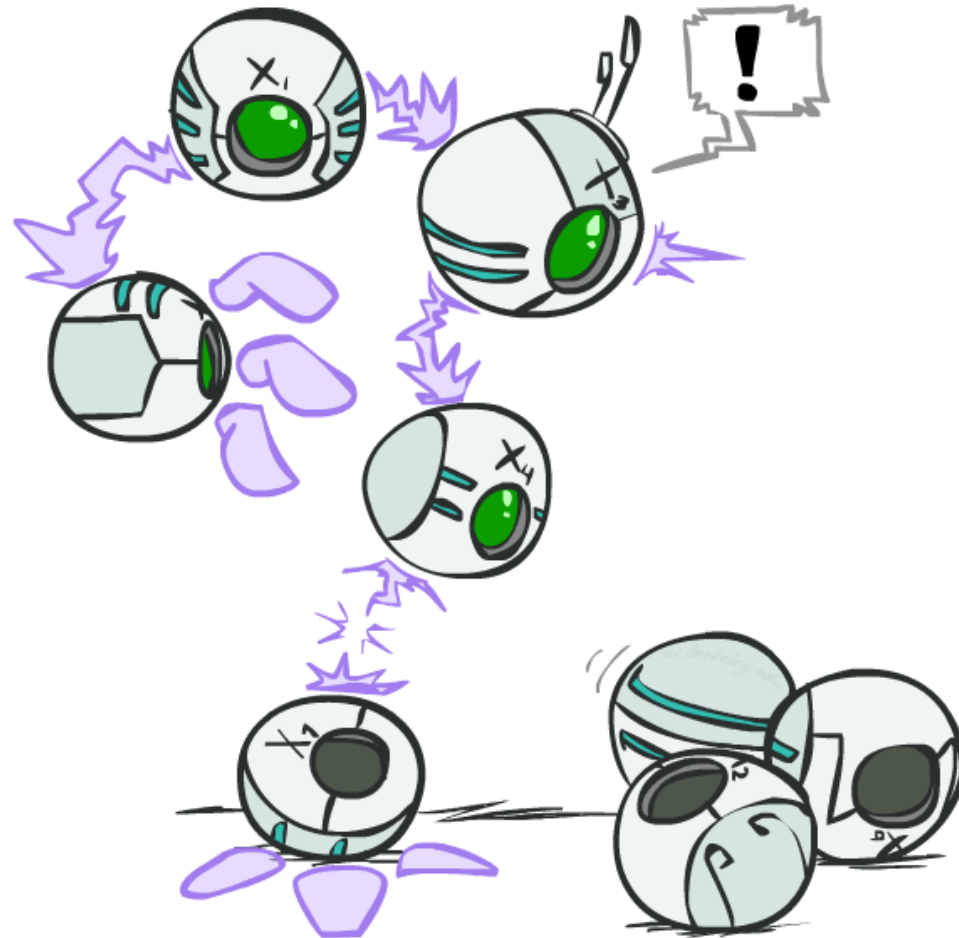
$P(L)$

L	
+l	0.134
-l	0.866

A factor of
exponential
size!



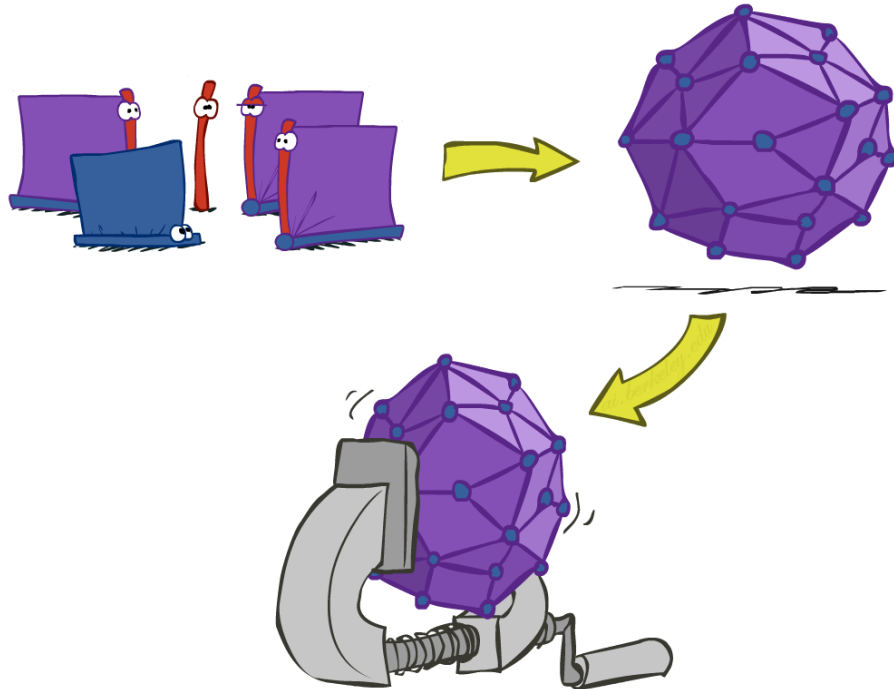
Variable Elimination



Inference by Enumeration vs. Variable Elimination

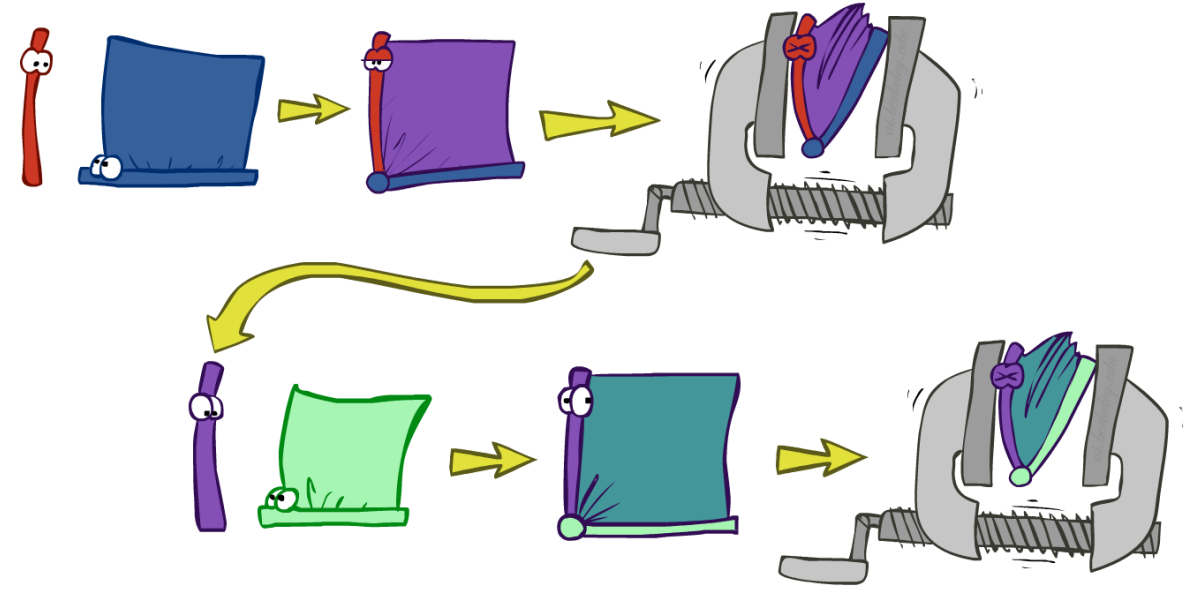
- Why is inference by enumeration so slow?

- You join up the whole joint distribution before you sum out the hidden variables



- Idea: interleave joining and elimination!

- Called “Variable Elimination”
- Still NP-hard, but usually much faster than inference by enumeration

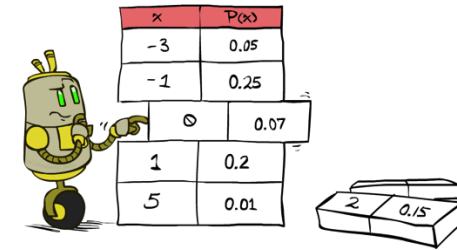


Variable Elimination

- Query: $P(Q|E_1 = e_1, \dots, E_k = e_k)$

- Start with initial factors:

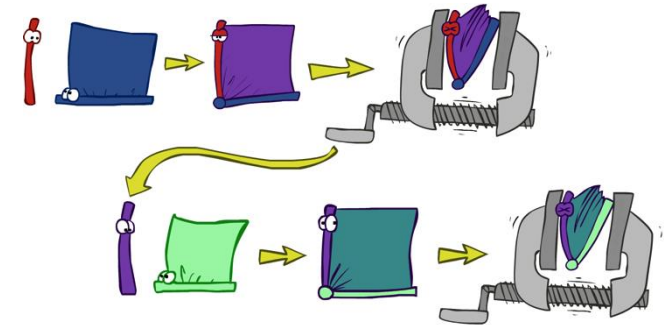
- Local CPTs (but instantiated by evidence)



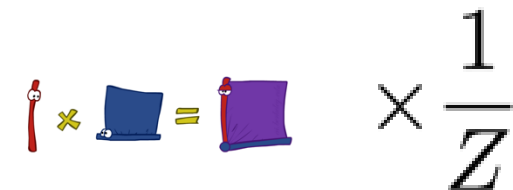
x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

- While there are still hidden variables (not Q or evidence):

- Pick a hidden variable H
 - Join all factors mentioning H
 - Eliminate (sum out) H

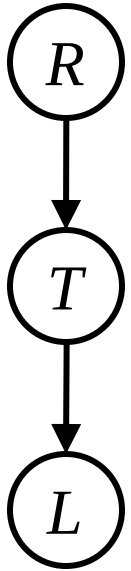


- Join all remaining factors and normalize



$$\text{stick figure} \times \text{blue square} = \text{purple square} \times \frac{1}{Z}$$

Traffic Domain



$$P(L) = ?$$

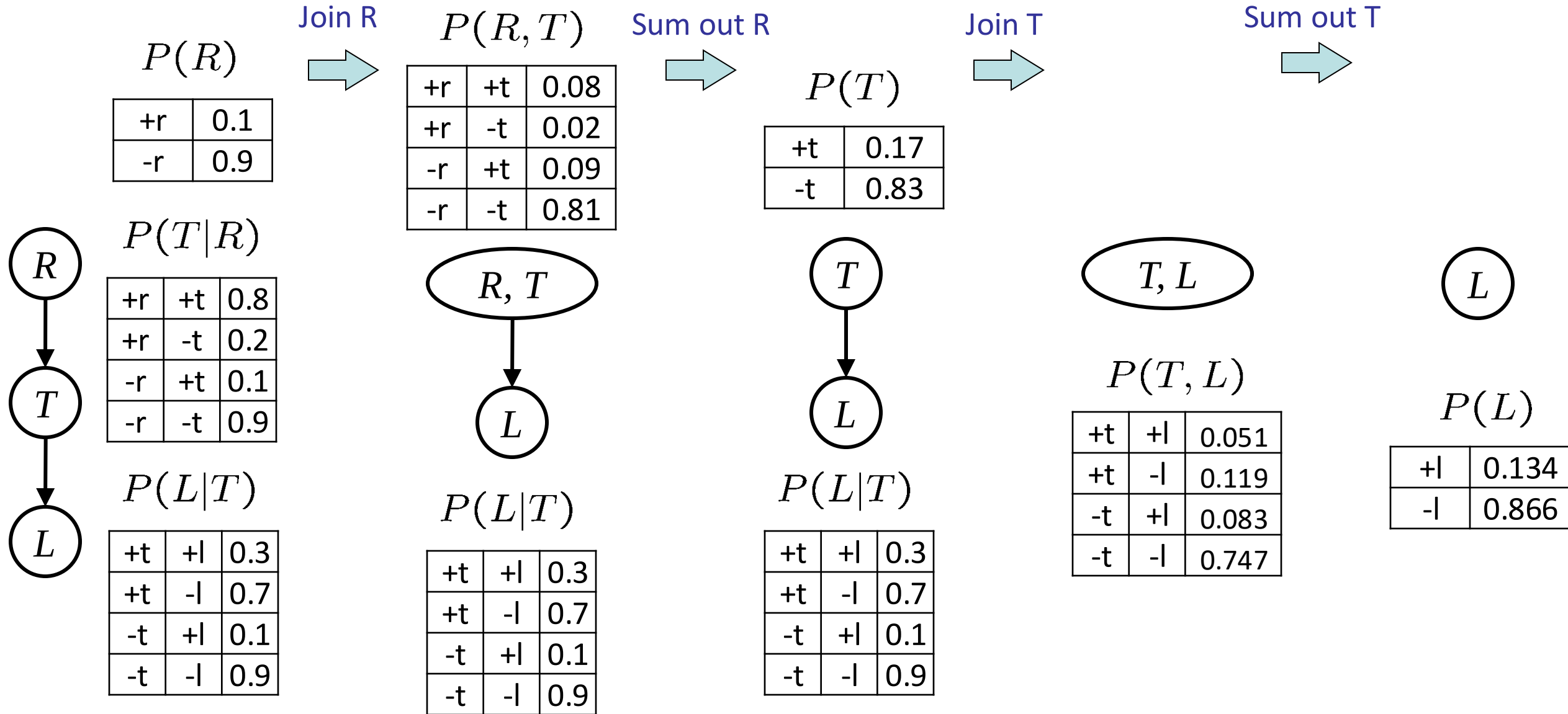
■ Inference by Enumeration

$$= \sum_t \sum_r \underbrace{P(L|t)P(r)P(t|r)}_{\text{Join on } r} \underbrace{}_{\text{Join on } t} \underbrace{}_{\text{Eliminate } r} \underbrace{}_{\text{Eliminate } t}$$

■ Variable Elimination

$$= \sum_t P(L|t) \underbrace{\sum_r P(r)P(t|r)}_{\text{Join on } r} \underbrace{}_{\text{Eliminate } r} \underbrace{}_{\text{Join on } t} \underbrace{}_{\text{Eliminate } t}$$

Variable Elimination



Evidence

- If evidence, start with factors that select that evidence

- No evidence uses these initial factors:

$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Computing $P(L|+r)$ the initial factors become:

$$P(+r)$$

+r	0.1
----	-----

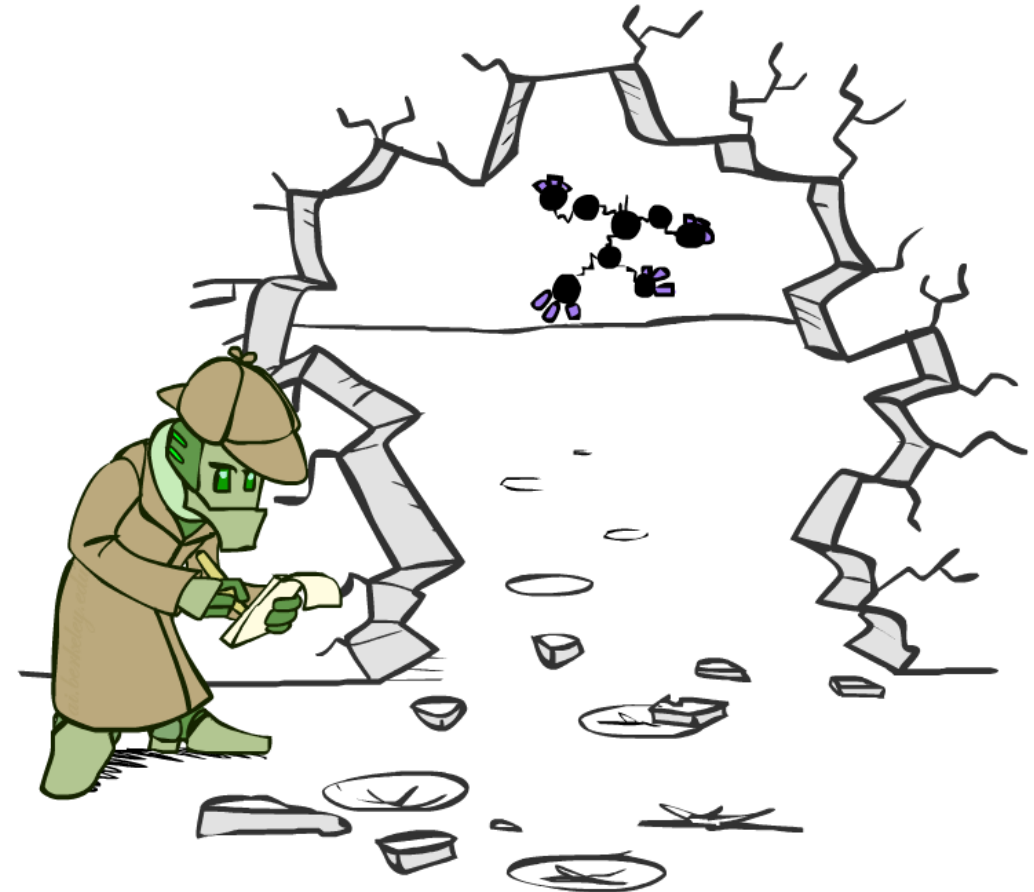
$$P(T|+r)$$

+r	+t	0.8
+r	-t	0.2

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- We eliminate all vars other than query + evidence



Evidence II

- Result will be a selected joint of query and evidence
 - E.g. for $P(L \mid +r)$, we would end up with:

$$P(+r, L)$$

+r	+l	0.026
+r	-l	0.074

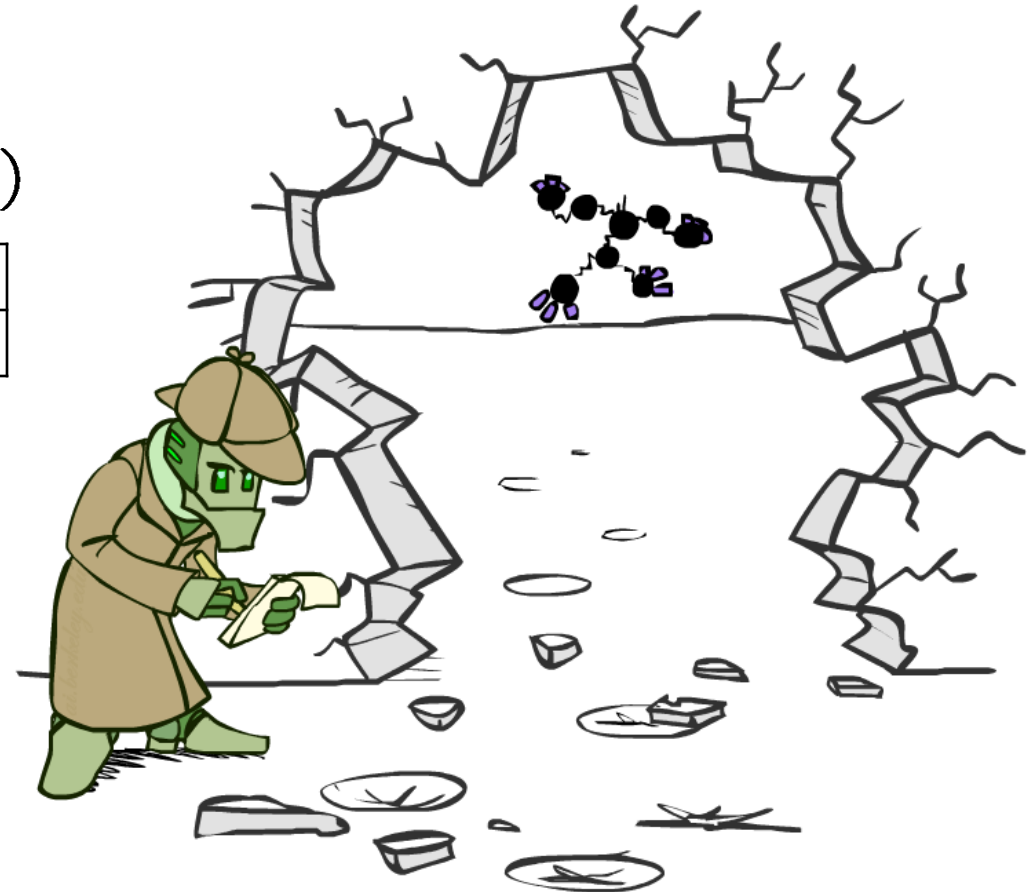
Normalize



$$P(L \mid +r)$$

+l	0.26
-l	0.74

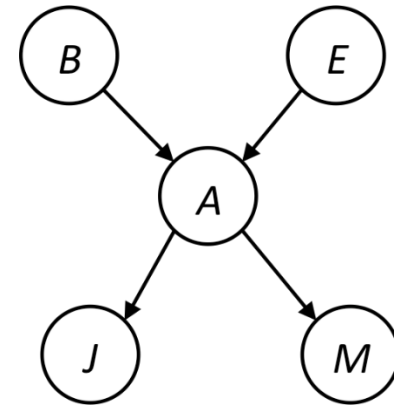
- To get our answer, just normalize this!
- That's it!



Example

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
--------	--------	-------------	----------	----------

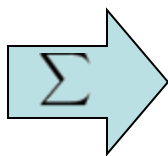
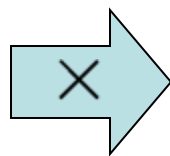


Choose A

$$P(A|B, E)$$

$$P(j|A)$$

$$P(m|A)$$



$$P(j, m|B, E)$$

$P(B)$	$P(E)$	$P(j, m B, E)$
--------	--------	----------------

Example

$P(B)$	$P(E)$	$P(j, m B, E)$
--------	--------	----------------

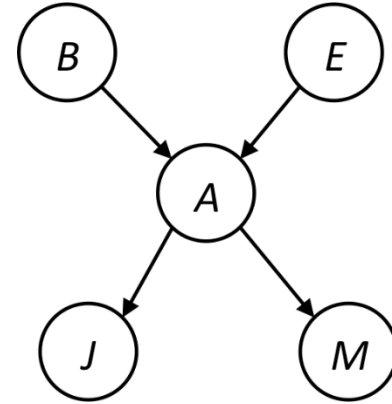
Choose E

$$\begin{array}{c} P(E) \\ P(j, m|B, E) \end{array} \xrightarrow{\times} \xrightarrow{\Sigma} P(j, m|B)$$

$P(B)$	$P(j, m B)$
--------	-------------

Finish with B

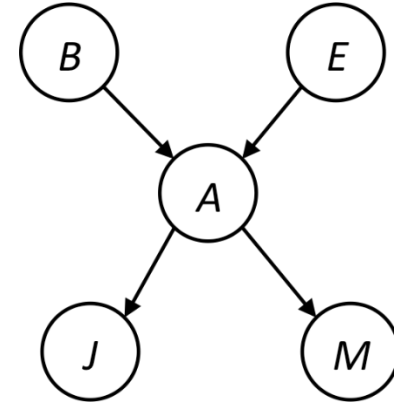
$$\begin{array}{c} P(B) \\ P(j, m|B) \end{array} \xrightarrow{\times} P(j, m, B) \xrightarrow{\text{Normalize}} P(B|j, m)$$



Same Example in Equations

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
--------	--------	-------------	----------	----------



$$\begin{aligned}
 P(B|j, m) &\propto P(B, j, m) \\
 &= \sum_{e, a} P(B, j, m, e, a) \\
 &= \sum_{e, a} P(B)P(e)P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e) \sum_a P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e)f_1(B, e, j, m) \\
 &= P(B) \sum_e P(e)f_1(B, e, j, m) \\
 &= P(B)f_2(B, j, m)
 \end{aligned}$$

marginal can be obtained from joint by summing out

use Bayes' net joint distribution expression

use $x*(y+z) = xy + xz$

joining on a, and then summing out gives f_1

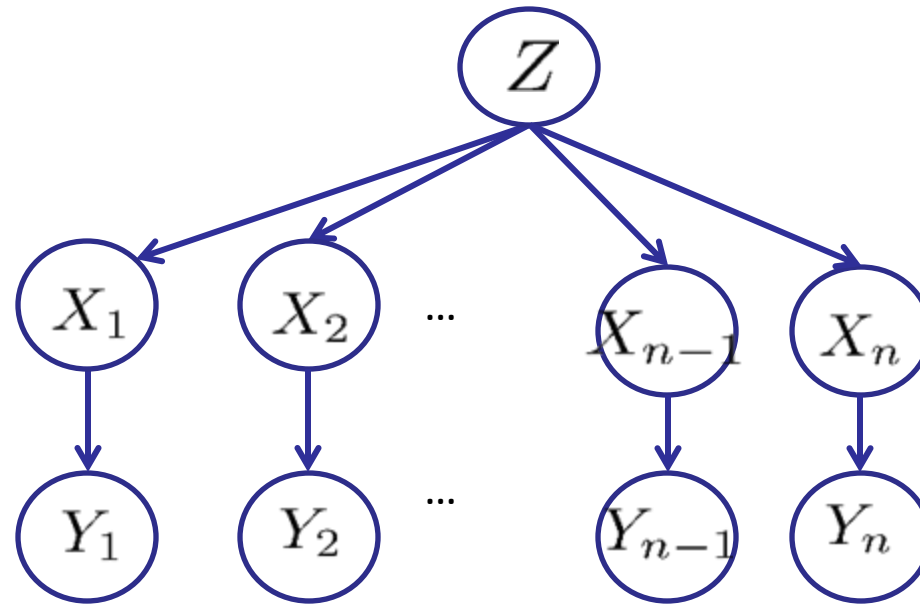
use $x*(y+z) = xy + xz$

joining on e, and then summing out gives f_2

All we are doing is exploiting $uwy + uwz + uxy + uxz + vwy + vwz + vxy + vxz = (u+v)(w+x)(y+z)$ to improve computational efficiency!

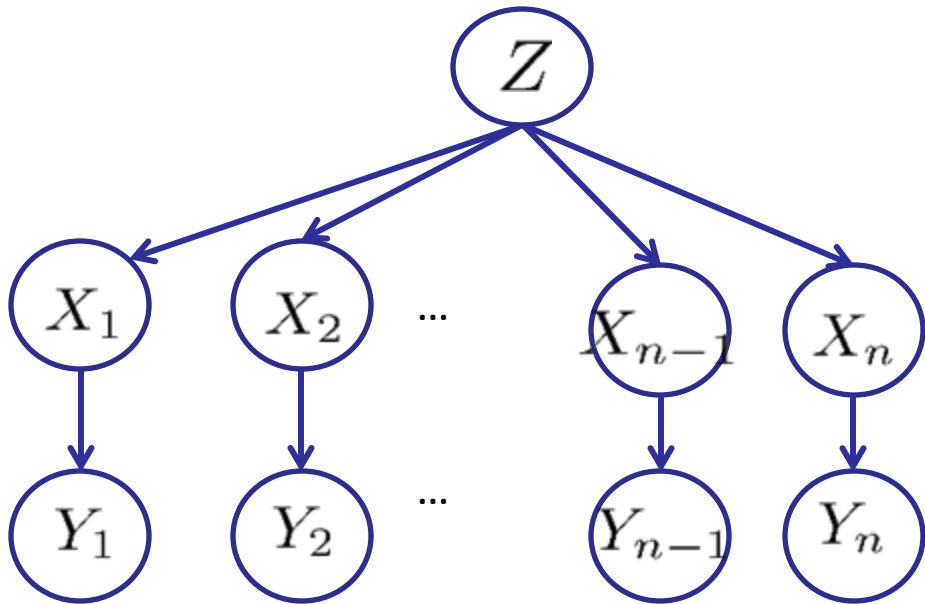
Variable Elimination Ordering

- Query: $P(X_n | y_1, \dots, y_n)$
- Two different orderings: $Z, X_1, \dots, X_{n-1}$ and $X_1, \dots, X_{n-1}, Z$.
- What is the size of the maximum factor generated for each of the orderings?



Variable Elimination Ordering

- $Z, X_1, \dots, X_{n-1}$



$$P(Z)P(X_1|Z)P(X_2|Z), \dots, P(X_n|Z)$$



$$f_1(X_1, X_2, \dots, X_n)$$



$$f_2(y_1, X_2, \dots, X_n)$$



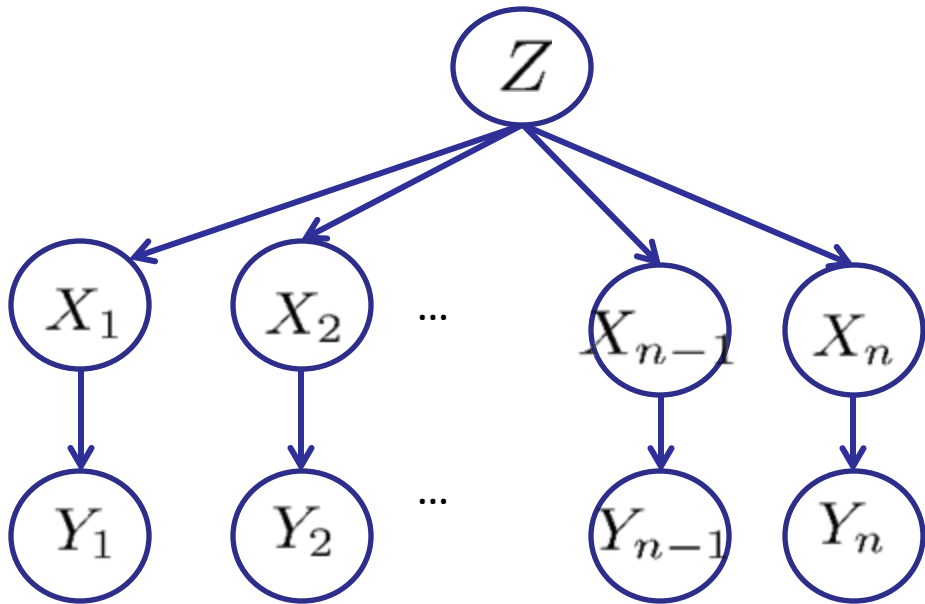
$$f_3(y_1, y_2, \dots, X_n)$$



$$2^{n+1}$$

Variable Elimination Ordering

- $X_1, \dots, X_{n-1}, Z$



$$P(X_1|Z)P(y_1|X_1)$$



$$f_1(Z, y_1)$$



$$f_1(Z, y_1), f_2(Z, y_2), \dots, f_{n-1}(Z, y_{n-1}), P(Z), P(X_n|Z)$$



$$f_n(X_n, y_1, \dots, y_{n-1})$$



2^2

VE: Computational Complexity

- The size of the largest factor determines the time and space complexity of VE
- The elimination ordering can greatly affect the size of the largest factor.
 - E.g., previous slide's example 2^{n+1} vs. 2^2
- Does there always exist an ordering that only results in small factors?
 - No!

Reduction from 3SAT

$$(x_1 \vee x_2 \vee \neg x_3) \wedge (\neg x_1 \vee x_3 \vee \neg x_4) \wedge (x_2 \vee \neg x_2 \vee x_4) \wedge (\neg x_3 \vee \neg x_4 \vee \neg x_5) \wedge (x_2 \vee x_5 \vee x_7) \wedge (x_4 \vee x_5 \vee x_6) \wedge (\neg x_5 \vee x_6 \vee \neg x_7) \wedge (\neg x_5 \vee \neg x_6 \vee x_7)$$

$$P(X_i = 0) = P(X_i = 1) = 0.5$$

$$Y_1 = X_1 \vee X_2 \vee \neg X_3$$

...

$$Y_8 = \neg X_5 \vee X_6 \vee X_7$$

$$Y_{1,2} = Y_1 \wedge Y_2$$

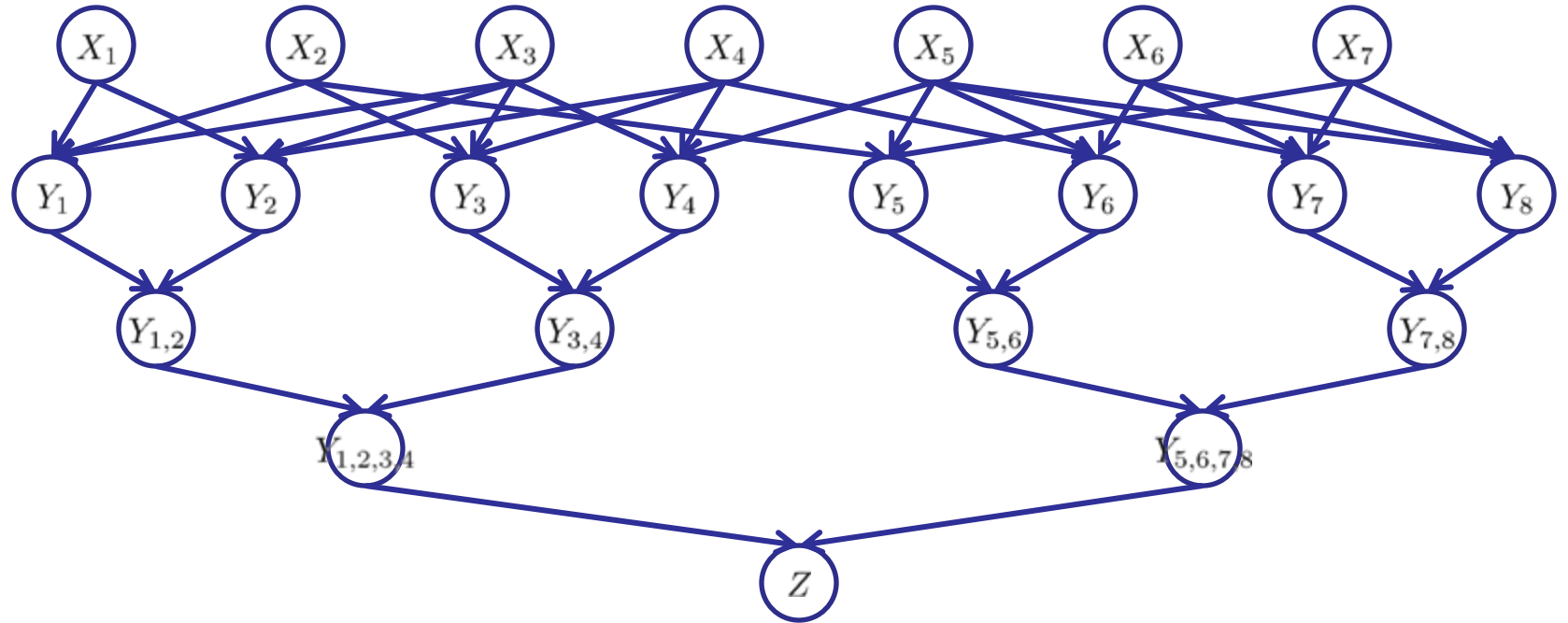
...

$$Y_{7,8} = Y_7 \wedge Y_8$$

$$Y_{1,2,3,4} = Y_{1,2} \wedge Y_{3,4}$$

$$Y_{5,6,7,8} = Y_{5,6} \wedge Y_{7,8}$$

$$Z = Y_{1,2,3,4} \wedge Y_{5,6,7,8}$$

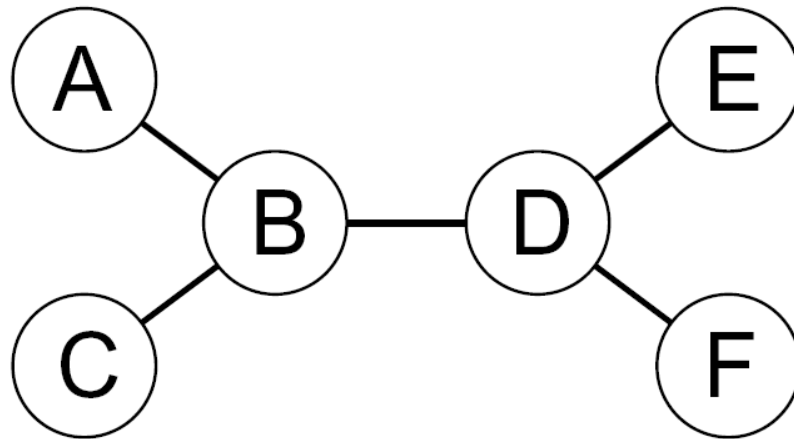


- If we can answer $P(z)$ equal to zero or not, we answered whether the 3-SAT problem has a solution.
- Hence inference in Bayes' nets is NP-hard. No known efficient probabilistic inference in general.

When do we have tractable inference?

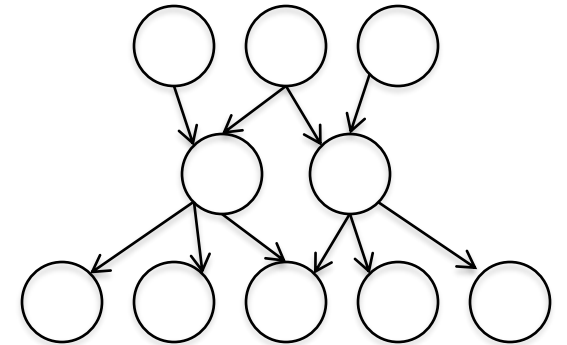
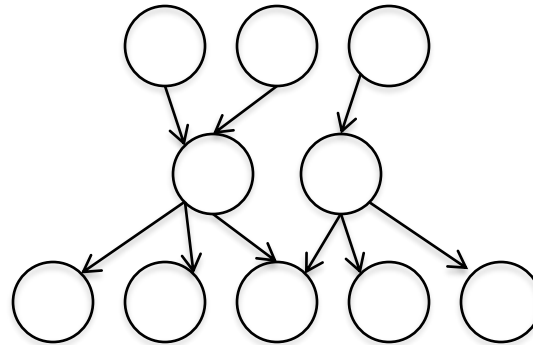
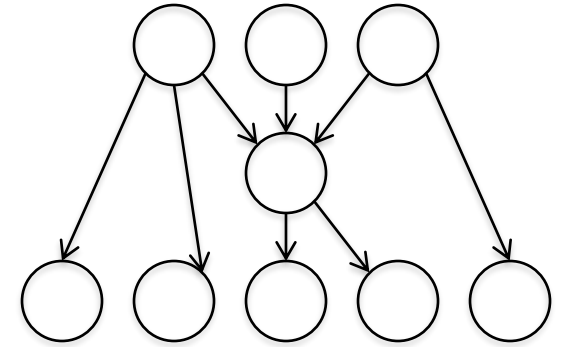
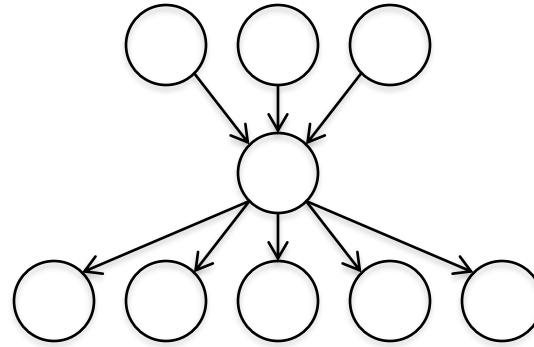
- Recall: Tree-Structured CSPs

- CSP is NP-hard in general
- If the constraint graph has no loops (i.e., tree), the CSP can be solved in linear time!



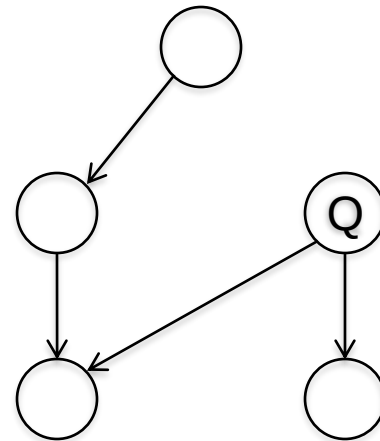
Polytrees

- A polytree is a directed graph with no undirected cycles

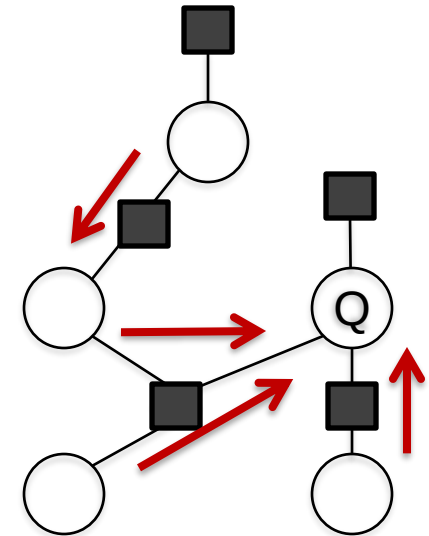


Variable Elimination on Polytrees

- For poly-tree BNs, the complexity of VE is *linear in the BN size* (number of CPT entries) with the following elimination ordering:
 - Convert to a factor graph
 - Take Q as the root
 - Eliminate from the leaves towards the root



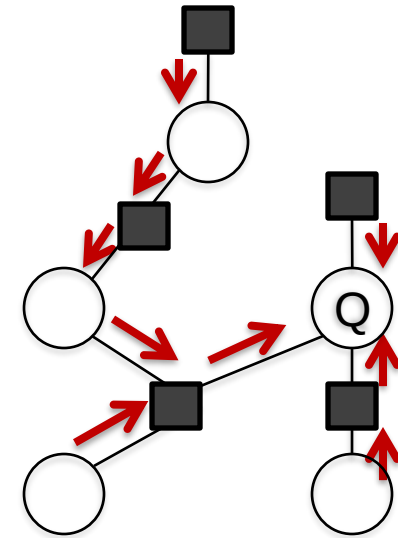
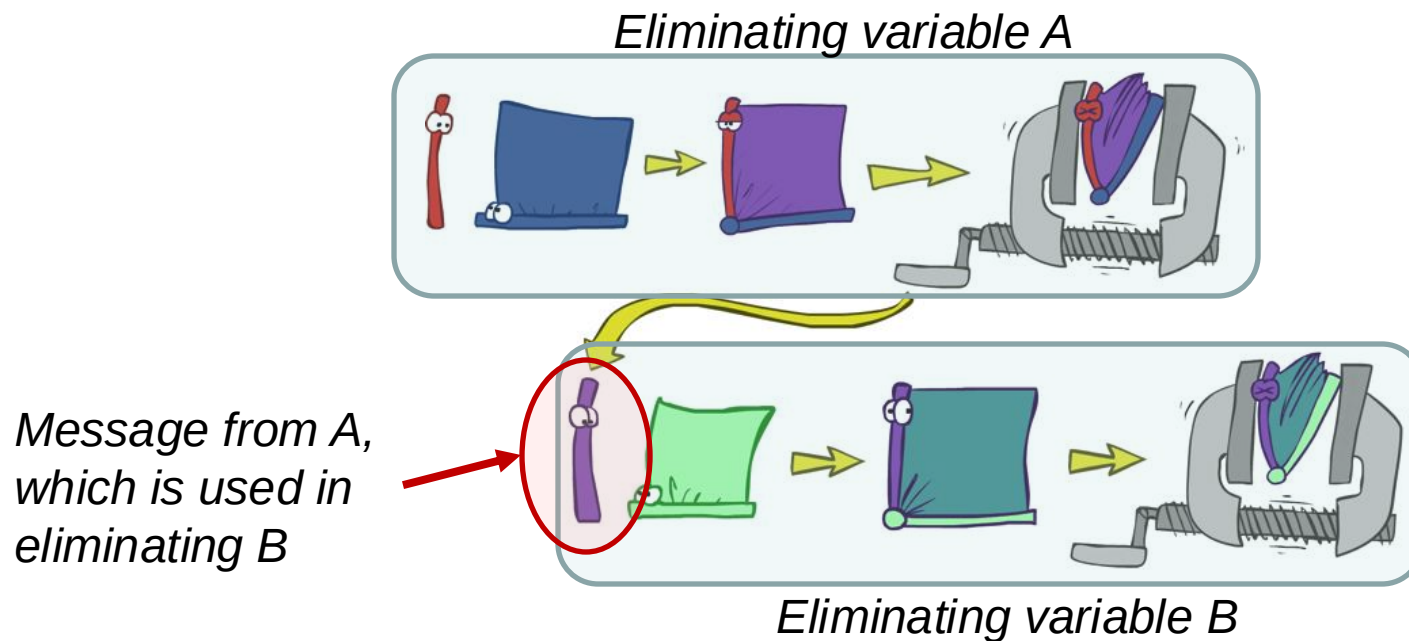
Bayesian Network



Factor Graph

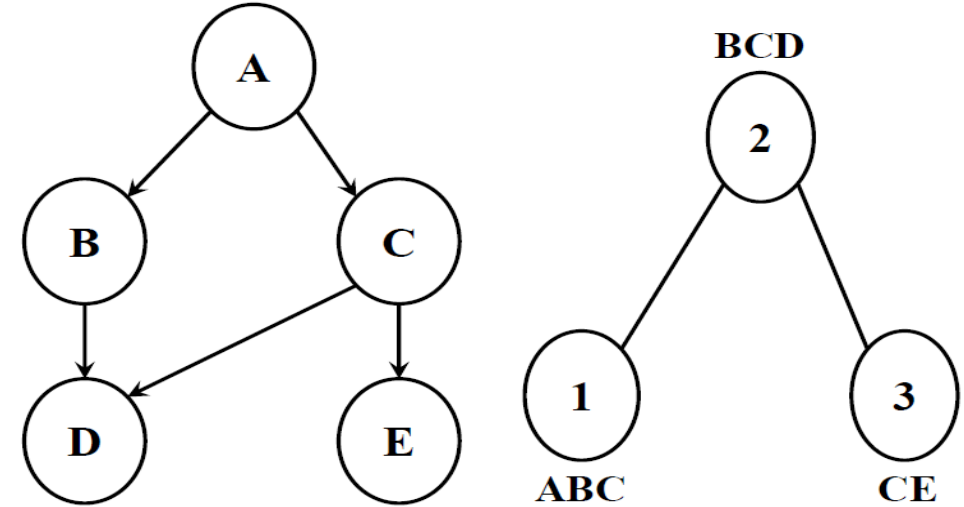
Variable Elimination on Polytrees

- VE for poly-tree BNs is equivalent to
 - Sum-product message passing algorithm or belief propagation algorithm (i.e., passing messages/beliefs from leaf nodes to the root node)



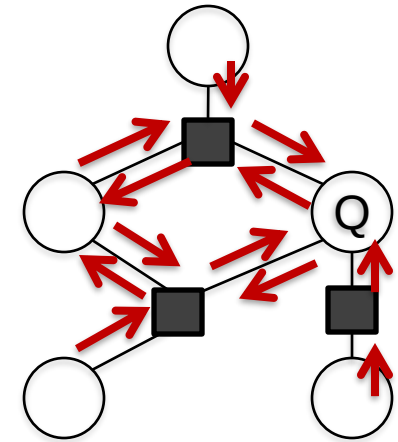
Message Passing on General Graphs

- Exact inference: Junction Tree Algorithm
 - Group individual nodes to form cluster nodes in such a way that the resulting network is a polytree (called a **junction tree** or **join tree**)
 - Run a sum-product-like algorithm on the junction tree.
 - Intractable* on graphs with large cliques (i.e., large **tree-width**).



Message Passing on General Graphs

- Approximate inference: Loopy Belief Propagation
 - Simply pass the messages on the general graph
 - Will not terminate with loops
 - Run until convergence (not guaranteed!)
 - *Approximate* but *tractable* for large graphs.
 - Sometime works well, sometimes not at all.



Summary

- Exact inference of Bayesian networks
 - Enumeration
 - exponential complexity
 - Variable Eliminating
 - worst-case exponential complexity, often better
 - VE on polytrees
 - linear complexity
 - = message passing
 - Message passing on general graphs
 - Junction Tree Algorithm
 - Loopy Belief Propagation: no longer exact